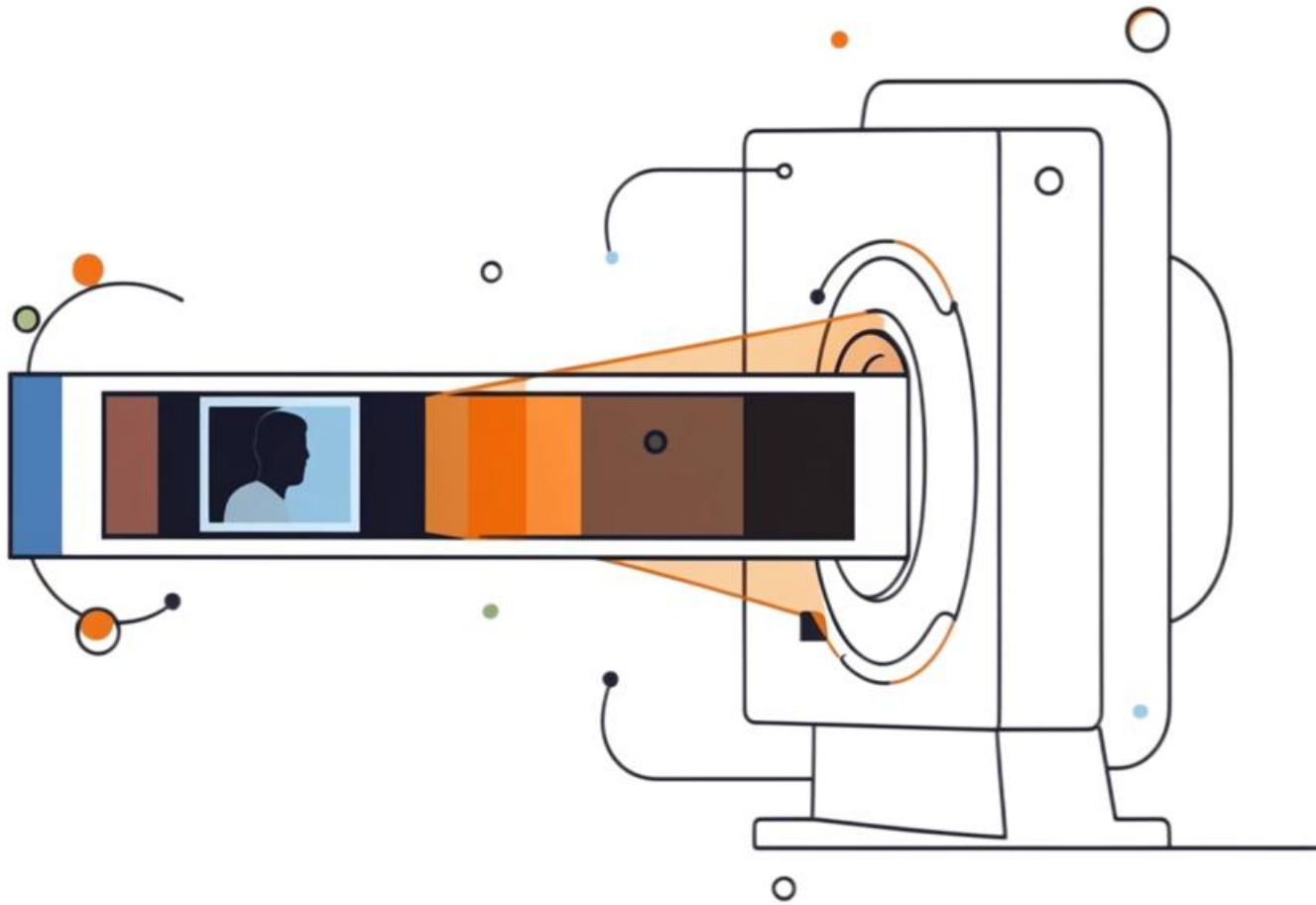
IMAGE PROCESSING · FOUNDATIONS

Image histograms

A visual guide to intensity, contrast, and practical OpenCV workflows.



A histogram turns an image into an intensity story

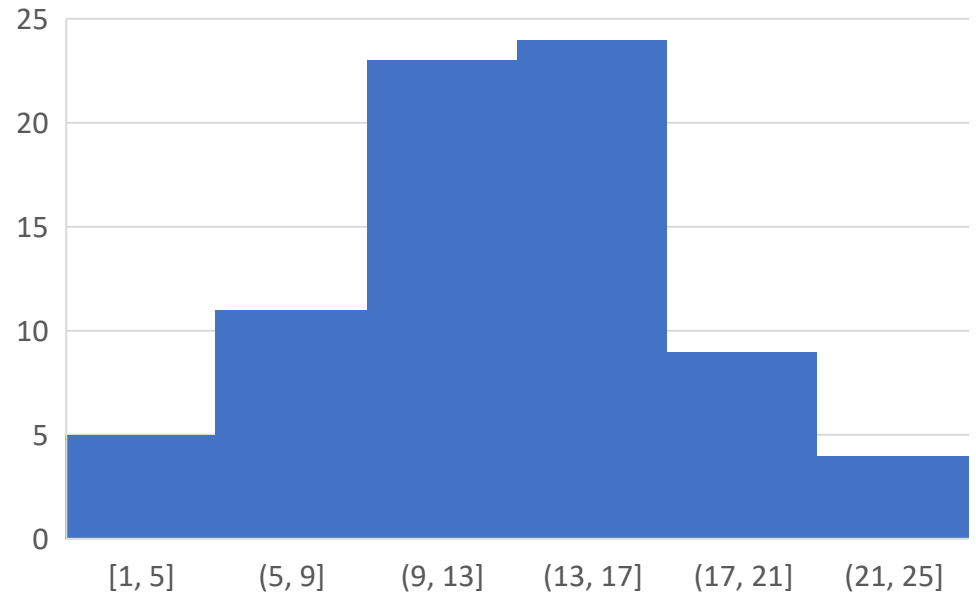
Every pixel contributes one value to a distribution. The histogram counts those values, turning a visual surface into something we can inspect, compare, and transform.

Pixel values

Intensity on the x-axis.

Count

Pixels at each value.



For an 8-bit grayscale image, the intensity range is commonly **0–255**.

02 · SHAPE AS EVIDENCE

Shape reveals darkness, brightness and contrast

Left · darker pixels

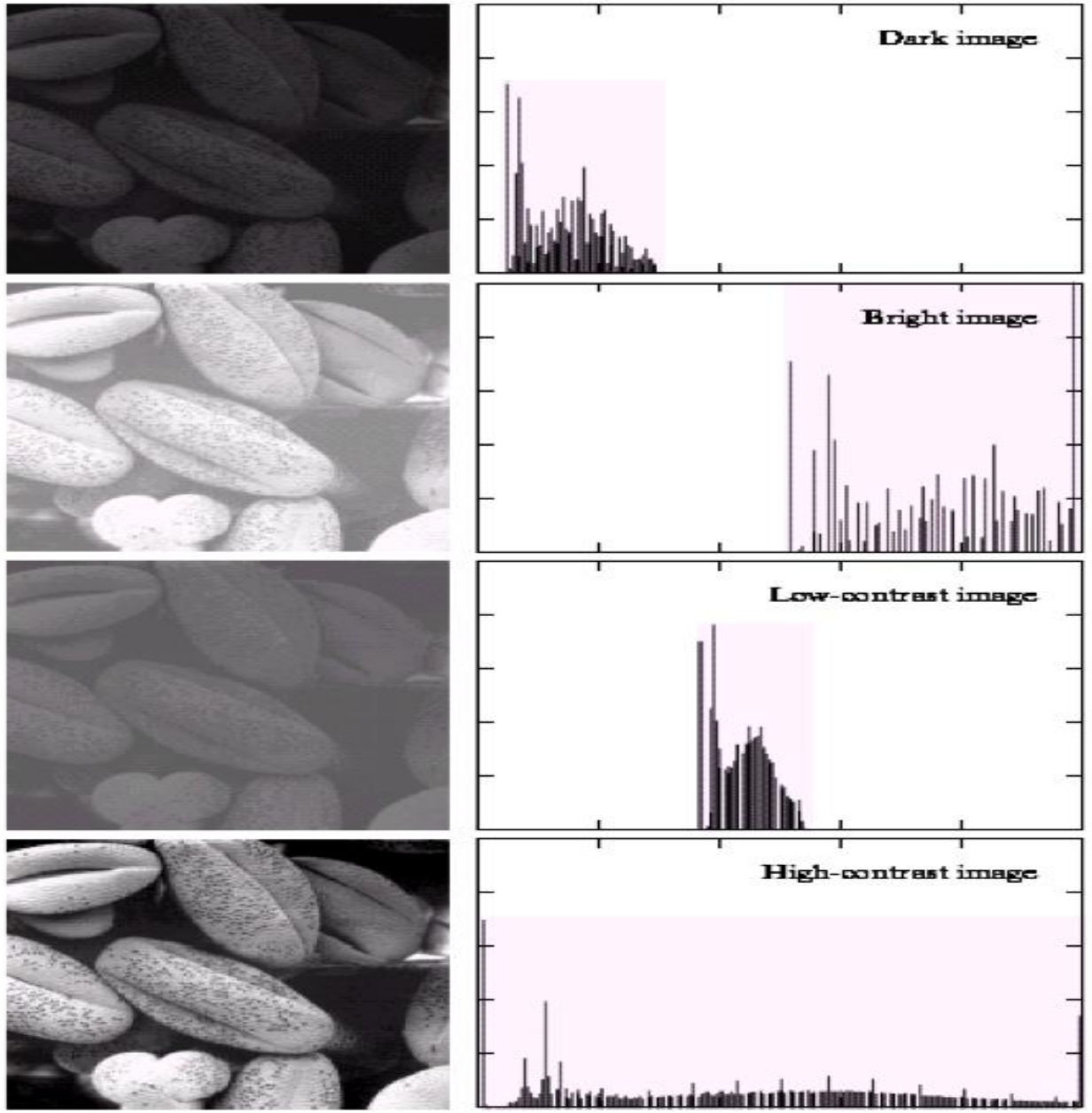
Middle · midtones

Right · brighter pixels

Widespread · Stronger contrast

Narrow spread · Low contrast

Peak height tells you how many pixels occupy an intensity range.



03 · COLOR CHANNELS

One image can have one histogram per channel

Grayscale gives one intensity distribution. A color image can be inspected through separate red, green, and blue channel histograms.

Channel views can expose color casts and clipping that a single grayscale view hides.



Grayscale

One distribution of intensity.



Red

Inspect the red contribution.



Green

Inspect the green contribution.



Blue

Inspect the blue contribution.

Equalization and matching solve different problems

Equalization

“Use the image’s own distribution to improve contrast.”

Best when the image needs a quick, parameter-light contrast enhancement.

Matching

“Make this image’s tonal distribution resemble that reference.”

Best when consistency with a reference image or collection matters.

Equalization asks, “How can this image use more of its range?” Matching asks, “Whose range should it resemble?”

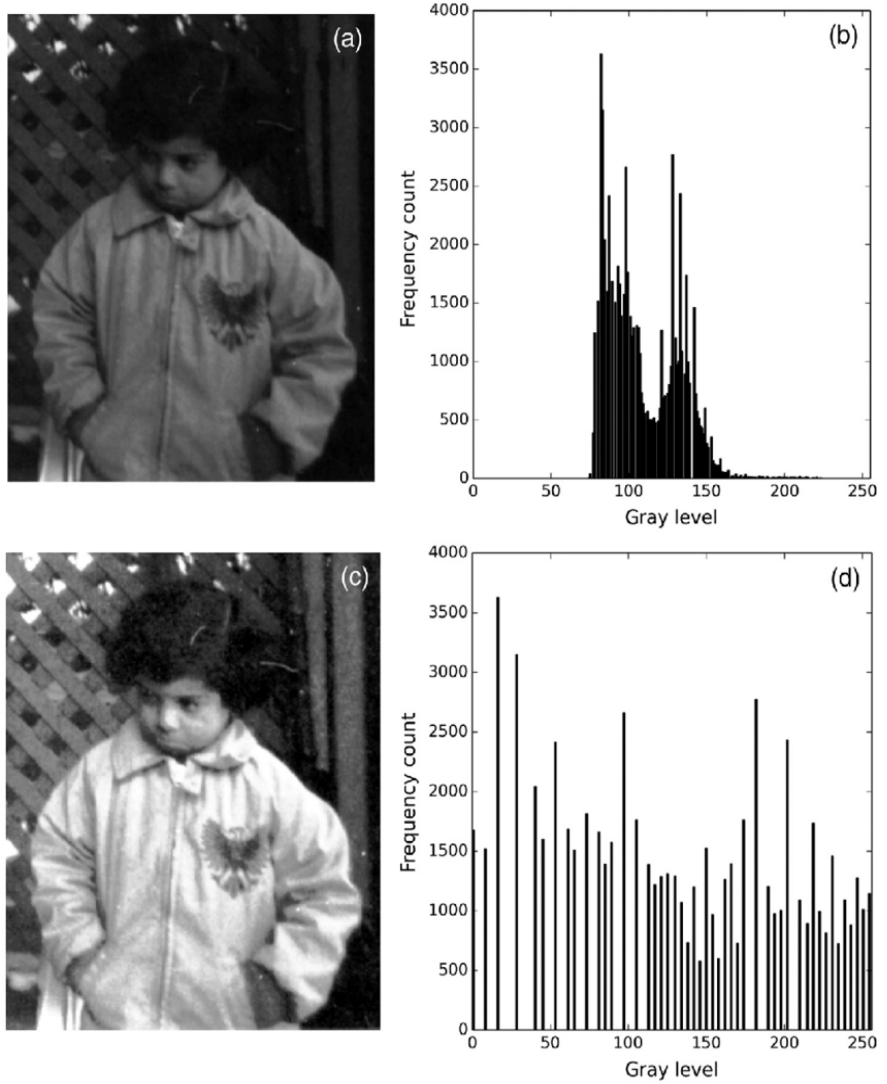
Histogram Equalization

Histogram equalization is a technique for adjusting image intensities to enhance contrast.
$p_r(r_k)$: The probability of occurrence of gray level r_k, where:

$$p_r(r_k) = \frac{n_k}{MN} \qquad k = 0, 1, 2, 3, \dots, L-1$$

This equation is histogram equalization or linearization. Thus, a processed (output) image is obtained by mapping each pixel with level r_k in the input image into a corresponding pixel with level s_k in the output image.

$$S_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) \qquad k = 0, 1, 2, 3, \dots, L-1$$



Histogram Equalization example

Assume that an image of 32x32 with 32 gray levels distributed as follow

Gray level	*12*	*14*	*15*	*16*	*18*	*19*	*20*	*21*	*22*
no. of pixels	*74*	*50*	*100*	*250*	*150*	*280*	*30*	*50*	*40*

- For the image represented in the previous table
- (i) Draw the image histogram
- (ii) Find the average(mean) intensity and the standard deviation
- (iii)Find the equalized histogram

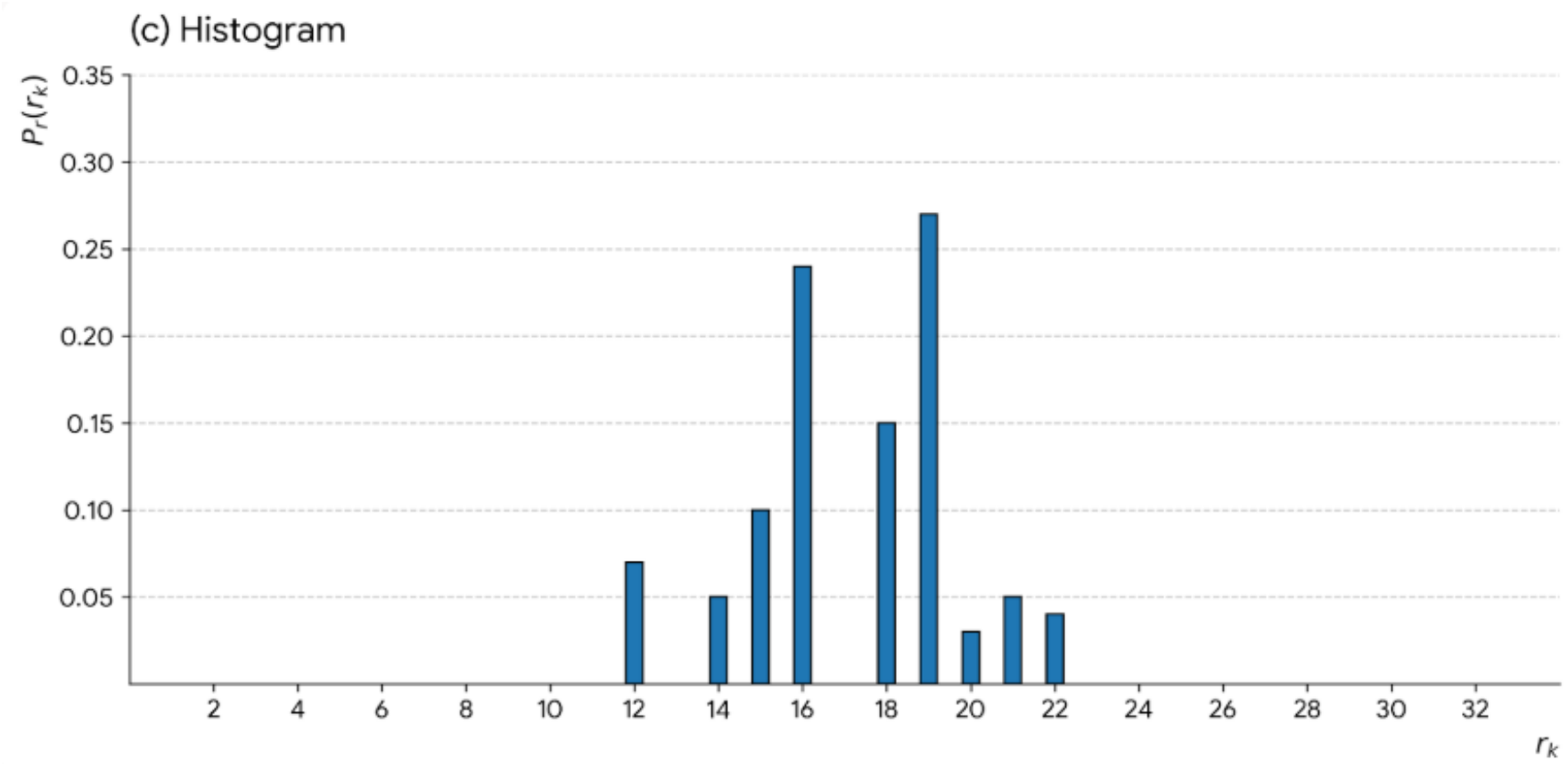
Histogram Equalization example solution

r_k	n_k	$P_r(r_k)=n_k/\mathrm{MN}$
12	74	74/1024=0.07
14	50	0.05
15	100	0.1
16	250	0.24
18	150	0.15
19	280	0.27
20	30	0.03
21	50	0.05
22	40	0.04

MN=32*32=1024

Histogram Equalization example solution cont.

(i) Draw the image histogram



Histogram Equalization example solution cont.

(ii) Find the average(mean) intensity and the standard deviation

The mean (Average intensity) of r:

$\mu = \sum_{i=0}^{L-1} r_i P(r_i)$ = (12*0.07)+(14*0.05)+(15*0.1)+…… = 17.24

Variance σ^2 :

$\sigma^2 = \sum_{i=0}^{L-1} (r_i - \mu)^2 P(r_i)$ =$(12 - 17.24)^2 * 0.07 + (14 - 17.24)^2 * 0.05 +$

$\ldots\ldots\ldots\ldots.. = 6.09$

standard deviation σ =$\sqrt{variance}$=$\sqrt{6.09}$=2.47

Histogram Equalization example solution cont.

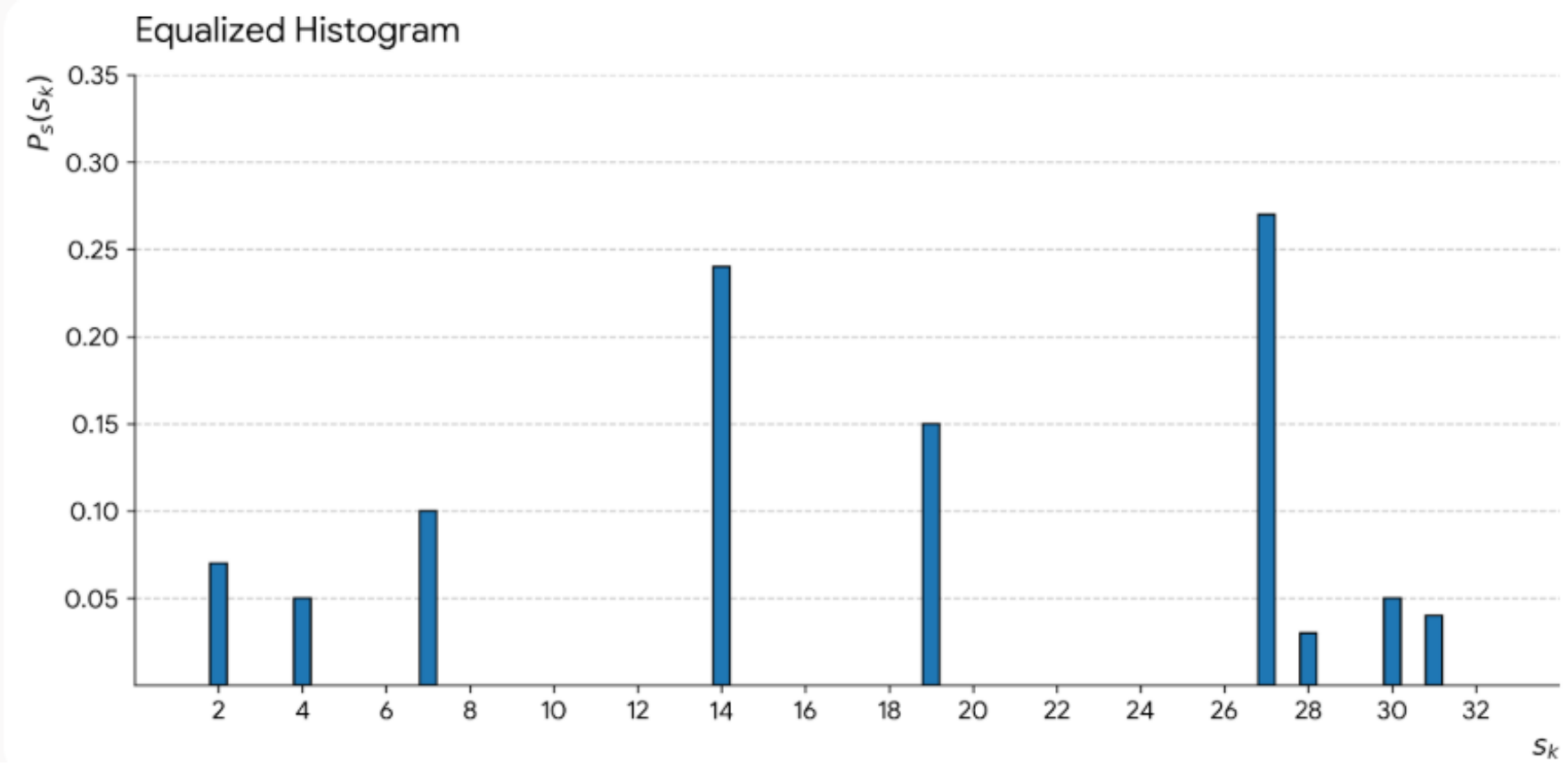
(iii)Find the equalized histogram

$$S_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) \qquad k = 0, 1, 2, 3, \dots, L-1 \qquad L-1 = 31$$

s_{12} =31*(0.07)=2.17 ≅ 2	P(s_2)=0.07
$s_{14} = 31 * (0.05 + 0.07) = 3.72 \cong 4$	P(s_4)=0.05
$s_{15} = 31 * (0.1 + 0.05 + 0.07) = 6.82 \cong 7$	P(s_7)=0.1
$s_{16} = 14.26 \cong 14$	P(s_{14})=0.24
s_{18} =18.91≅ 19	P(s_{19})=0.15
s_{19} =27.28≅ 27	P(s_{27})=0.27
s_{20}=28.21≅ 28	P(s_{28})=0.03
s_{21} =29.76≅ 30	P(s_{30})=0.05
s_{22} =31	P(s_{31})=0.04

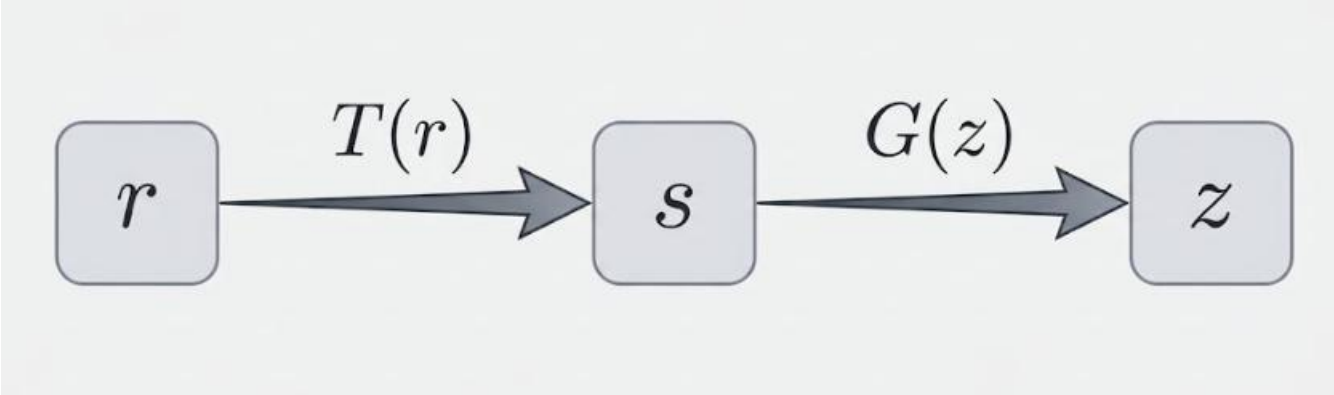
Histogram Equalization example solution cont.

(iii)Find the equalized histogram



Histogram Matching

Histogram matching is the method used to generate a processed image that has a specified histogram.



$$G(z_q) = (L-1)\sum_{i=0}^{q} p_z(z_i) \qquad q = 0, 1, 2, 3, \ldots, L-1$$

Histogram Matching example

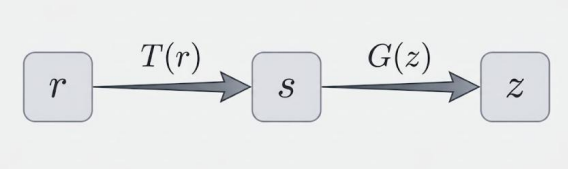
For the previous example:

(iv) Find the specified histogram according the following values?

Z_q	4	6	8	10	12	16	18	20	22
Specified $P_z(z_q)$	0.05	0.1	0.1	0.15	.2	0.15	0.1	0.1	0.05

Histogram Matching example solution

(iv) Find the specified histogram



$$G(z_q) = (L-1)\sum_{i=0}^{q} p_z(z_i)$$

$G(z_4) = 31*0.05=1.55 \cong 2$
$G(z_6) = 31*(0.1+0.05)=4.56 \cong 5$
$G(z_8) =31*(0.1+0.1+0.05)=7.75 \cong 8$
$G(z_{10}) =12.4 \cong 12$
$G(z_{12}) =18.6 \cong 19$
$G(z_{16}) =23.25 \cong 23$
$G(z_{18}) =26.35 \cong 26$
$G(z_{20}) =29.45 \cong 29$
$G(z_{22}) =31$

Histogram Matching example solution cont.

(iv) Find the specified histogram

s_k	G(z_q)	z_q(actual)	P(z_q)
2	2	4	0.07
4	5	6	0.05
7	8	8	0.1
14	12	10	0.24
19	19	12	0.15
27	26	18	0.27
28	29	20	0.03+0.05=0.08
30	29	20	
31	31	22	0.04


Thank
You